

Enhanced YOLOv8 network with Extended Kalman Filter for wildlife detection and tracking in complex environments

Langkun Jiang, Li Wu^{*}

School of Computer Technology and Application, Qinghai University, Xining 810016, China

ARTICLE INFO

Keywords:

YOLOv8

Target detection

Kalman filter

Deformable convolution

Tracking

ABSTRACT

Amid a growing global focus on ecological conservation and biodiversity monitoring, the efficient identification and tracking of wildlife are essential for environmental research, wildlife protection, and habitat management. Nevertheless, intricate landscapes, varied animal sizes, and obstructions obstruct wildlife detection and tracking. This study introduces the wildT-YOLOv8n model, specifically engineered for the effective identification and tracking of animals. Initially, the Stable Diffusion model augments the dataset, establishing a basis for training data. Subsequently, enhancements to the YOLOv8n model are implemented through the incorporation of the deformable convolutional network DConv3 and the utilization of the C2f-DConv3 layer to augment feature extraction efficacy, while addressing detection challenges associated with small targets and intricate backgrounds by integrating the EMGA attention mechanism and the ASPFC feature fusion module. Enhancing the Extended Kalman Filter algorithm guarantees reliable and precise tracking. The research findings reveal that the wildT-YOLOv8n model attained an average detection accuracy (mAP50) of 88.54 % on the custom dataset, reflecting a 4.57 % enhancement over the original YOLOv8n model; the refined Extended Kalman Filter realizes a Multi-Object Tracking Accuracy (MOTA) of 40.35 %, representing a 3.923 % advancement over the original Kalman Filter. The results indicate the feasibility of accurately detecting and monitoring wildlife in intricate environments, offering significant insights for ecological research and biodiversity conservation, and aiding in the protection of endangered species.

1. Introduction

The preservation of ecological conservation and biodiversity has emerged as a significant challenge for contemporary society on a worldwide scale (Rawat and Agarwal, 2015). Wildlife, as a crucial element of ecosystems, immediately indicates the health and stability of the ecological environment (Upreti, 2023). Consequently, the precise and effective identification and monitoring of wildlife are crucial for environmental research, biodiversity preservation, natural reserve administration, and disease surveillance (Urbano et al., 2010). Advancements in technology, especially in artificial intelligence (AI) and computer vision, have enabled the development of automated tools for real-time wildlife monitoring and behavioral analysis. It markedly enhances monitoring efficiency, diminishes human resource expenditure, and delivers more comprehensive and precise data support, so furnishing a scientific foundation for ecological conservation decisions (Petso et al., 2022; Pimm et al., 2015). Nonetheless, the intricate and dynamic natural environment, species diversity, and morphological

variations among individuals present considerable obstacles to the automated detection and tracking of wildlife, necessitating the development of more efficient and precise detection and tracking technologies to meet the increasing demands of ecological monitoring (Kumar and Singh, 2017).

Conventional wildlife target detection techniques mostly encompass manual surveys and acoustic detection systems. Manual surveys represent a traditional method of wildlife inquiry, typically entailing the gathering of animal specimens or traces via transects, quadrats, and similar techniques for statistical analysis. This strategy necessitates considerable human resources and may disrupt the animals' habitat (Chandler and Royle, 2013; Gor et al., 2017). Conversely, sound-based detection systems, while able to identify the presence of animals to a degree, possess restricted information-gathering capabilities due to factors such as sound propagation range and surrounding noise (Gibb et al., 2019). In recent years, propelled by improvements in deep learning technology, researchers have developed many image-based techniques for wildlife detection and recognition to address the issues

^{*} Corresponding author.

E-mail address: wuli777@qhu.edu.cn (L. Wu).

<https://doi.org/10.1016/j.ecoinf.2024.102856>

Received 26 April 2024; Received in revised form 13 October 2024; Accepted 13 October 2024

Available online 16 October 2024

1574-9541/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

in this domain (Christin et al., 2019). Okafor et al. developed a classification model that integrates grayscale and color information with convolutional neural networks, effectively accomplishing animal categorization tasks (Okafor et al., 2016). Norouzzadeh et al. employed the AlexNet, GoogLeNet, VGG, and ResNet series of network models to successfully perform wildlife species detection and enumeration (Norouzzadeh et al., 2018). Trammell et al. utilized various convolutional neural networks for animal classification on the iNaturalist dataset, attaining positive results (Trammell et al., 2019). Tang et al. introduced a semi-supervised learning approach that proficiently categorized targets via feature extraction utilizing convolutional neural networks and the transfer of visual and semantic knowledge (Tang et al., 2016). Liu Wendong et al. suggested a dual-channel automatic wildlife recognition approach utilizing global-local features for detection (Wending et al., 2018). They delineated animal regions in infrared detection photos employing DarkNet-19 and utilized VGG16's dual-channel convolutional neural network for feature extraction and fusion, attaining elevated accuracy in wildlife area detection and recognition. Karp et al. effectively identified small and elusive species with thermal imaging technology and wildlife detection dogs, offering qualitative evaluations of the merits and drawbacks of these techniques in wildlife detection (Karp, 2020). Hamilton et al. proposed the integration of machine learning with drone-captured thermal imaging in intricate landscapes to tackle the issue of koala detection (Hamilton et al., 2020). Zhao et al. presented an improved YOLO network for real-time wildlife detection in the Yunnan Tongbiguan Nature Reserve, located in the southwestern tropical region of China (Zhao et al., 2021). They used the lightweight MobileNet architecture to supplant the YOLO backbone feature extraction network, facilitating swift detection on CPU platforms. Zhong et al. developed a deep learning network based on YOLO for the real-time detection of marine fauna in coral reef ecosystems, evaluating several methodologies and validating their efficacy in rapidly and accurately identifying marine species inhabiting coral reefs (Zhong et al., 2022). Jansi Rani et al. proposed a non-invasive method for measuring fish biomass in murky conditions by employing a modified YOLOv8 model for object detection and regression models such as Extra Trees Regressor, with excellent accuracy on the Aquatic Weight-Net dataset for the GIFT Tilapia species (Rani et al., 2024). Wagner et al. devised a technique utilizing colorized point clouds from Personal and Terrestrial Laser Scanning (PLS, TLS) in conjunction with pictures to autonomously identify marked trees and discern numbers on them, attaining excellent precision with a YOLOv8 model for number classification (Wagner et al., 2024). Cusick et al. devised a machine learning methodology utilizing the YOLOv6L model and Remotely Piloted Aircraft Systems (RPAS) to precisely enumerate Antarctic shag nests, attaining an F1 score over 0.95, thereby facilitating effective large-scale surveillance of seabird colonies in Antarctica (Cusick et al., 2024). Liu et al. introduced YOLO v8-SIM, an optimized camouflage detection model for *Alligator sinensis*, employing ResNet-18 alongside Biformer and Reverse Attention to augment detection accuracy in intricate environments by concentrating network attention and enhancing scale generalization via the Inner-IoU loss (Liu et al., 2024). Chen et al. presented YOLO-SAG, an enhanced wildlife detection algorithm derived from YOLOv8n, designed to optimize accuracy and speed in wildlife monitoring activities (Chen et al., 2024). Their technique was developed to tackle challenges such as the intricacy of outdoor situations and the diversity of animal sizes that impede detection accuracy. Roy et al. presented WilDect-YOLO for accurate automated identification and localization of endangered fauna (Roy et al., 2023). The model integrates DenseNet blocks with the CSPDarknet53 backbone network, employs SPP and a modified PANet, and attains effective detection performance in intricate situations. Nonetheless, these solutions continue to encounter certain challenges: Detection of diminutive targets - many animals exhibit significant size variation, with certain species, such as birds or small mammals, being minuscule and consequently prone to being overlooked or misunderstood in photographs due to their

diminutive dimensions. Complex backgrounds present a challenge; in natural environments, animals frequently camouflage with their surroundings, or the background is variable and cluttered, complicating algorithms' ability to accurately distinguish animals from the background, resulting in diminished recognition accuracy or misidentification of the background as the target.

The YOLO series algorithms have received significant interest in animal monitoring and tracking due to their exceptional efficiency in real-time performance and accuracy. YOLOv8 presented the BoT-SORT tracking algorithm for object tracking, which integrates efficient bounding box matching with motion estimates to provide continuous target tracking. The BoT-SORT algorithm employs a Kalman filter module to assess the target's motion condition and forecast its subsequent position (Aharon et al., 2022). The Kalman filter module enhances the estimation of the target state utilizing the system's dynamic model and measurement data, hence augmenting tracking accuracy and stability; yet, it possesses certain limitations (Simon, 2010). The Kalman filter module is predicated on a linear dynamic model and the assumption of Gaussian noise, which may yield suboptimal performance in nonlinear systems or scenarios involving non-Gaussian noise. The accuracy of the system's dynamic model, essential for the Kalman filter, must be high; departures from the actual situation may result in biased filtering outcomes. This paper presents the Extended Kalman Filter (EKF) to overcome the limitations of the Kalman filter module. The Extended Kalman Filter (EKF) linearizes nonlinear systems, so partially addressing the limitation of the Kalman filter, which is confined to linear systems. In contrast to conventional Kalman filtering, the Extended Kalman Filter exhibits superior adaptability and robustness in managing nonlinear systems and non-Gaussian noise. Nonetheless, the Extended Kalman Filter possesses several limitations (Garcia et al., 2019). The linearization process may create approximation errors, particularly in highly nonlinear systems, thus compromising the filter's efficacy and resulting in suboptimal outcomes. The Extended Kalman Filter necessitates great precision in the system's dynamic model, which might present difficulties in modeling and parameter optimization in complicated wildlife monitoring contexts. In practical applications, animals may hinder one another or be impeded by trees, bushes, etc., resulting in the loss of tracking targets, necessitating further enhancement of the robustness of the Extended Kalman Filter.

This work enhances the YOLOv8n model by concentrating on four elements: data augmentation, the incorporation of deformable convolutions, the implementation of attention mechanisms and feature fusion modules, and the refinement of the Extended Kalman Filter. The principal innovations are as follows:

- (1) Employing Stable Diffusion Data Augmentation: The introduction of the Stable Diffusion model to augment the original dataset establishes a more robust data foundation for model training, hence enhancing the model's generalization capability and resilience.
- (2) Improving the YOLOv8n Model: The YOLOv8n model incorporates the deformable convolution network DCNV3 in its head network, with the C2f.DCNV3 layer supplanting the conventional C2f layer, thereby markedly improving the model's ability to discern wildlife traits and augmenting its detection and recognition efficacy in intricate environments. The Efficient Multi-Scale and Global Attraction Mechanism (EMGA) attention mechanism is introduced, integrating two distinct attention mechanisms (EMA and GAM) to improve the efficacy of deep neural networks in computer vision applications. This method assists the model in prioritizing essential information, hence enhancing the accuracy and reliability of detection and tracking. This study enhances the model's detection efficacy for detecting compact targets amidst complicated backdrops, hence rendering it more applicable for real-world wildlife monitoring settings, based on the ASPFC module.

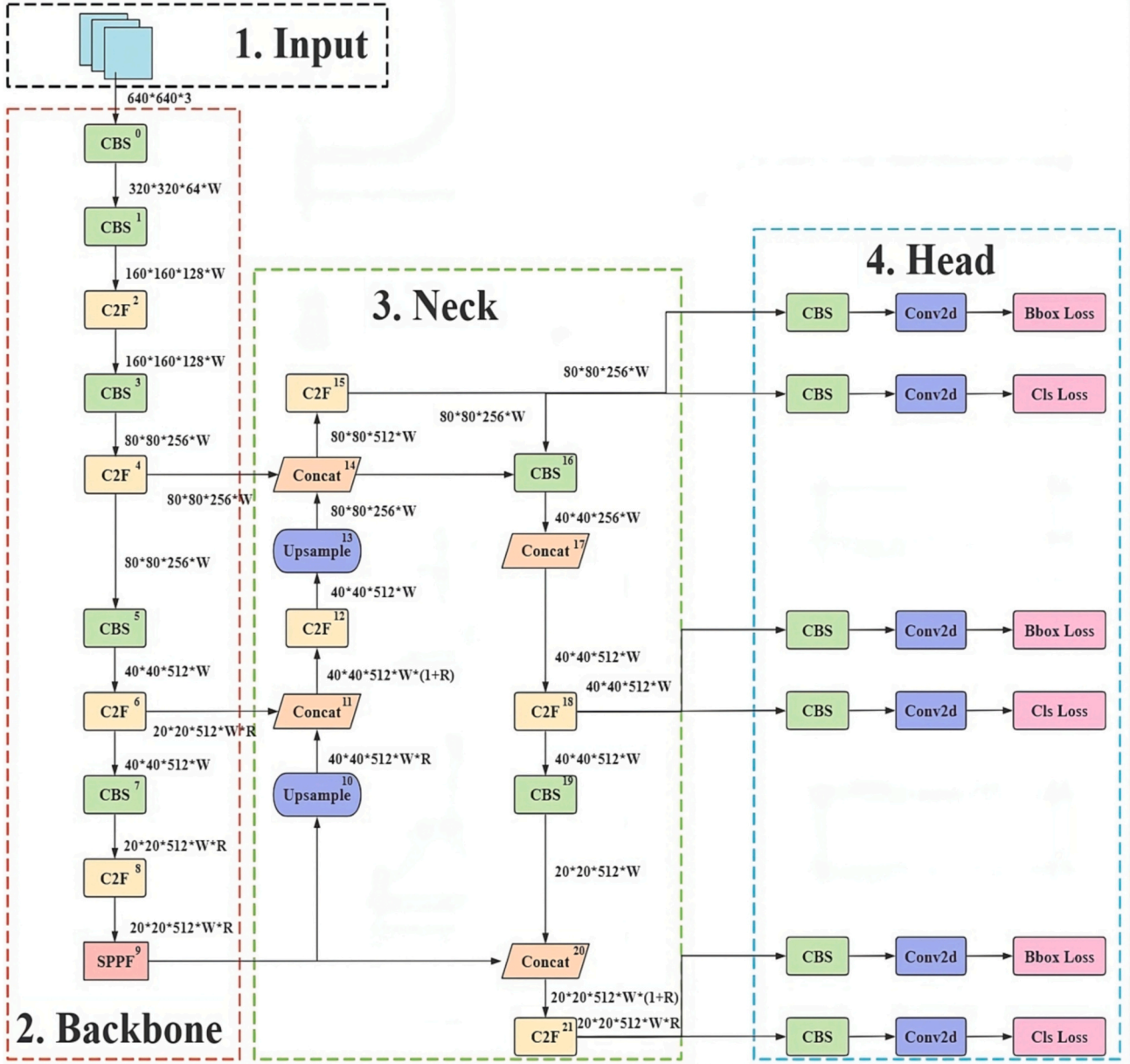


Fig. 1. YOLOv8 Network Structure.

- (3) This study enhances the Extended Kalman Filter algorithm to mitigate target loss due to obstructions. By optimizing the thresholding technique for the measurement-prediction residual during the measurement update process, the resilience and enhancement of the multi-target tracking system are achieved. It enhances the efficacy of wildlife tracking by tackling the issue of tracking impediments.

The organization of the paper is as follows: [Section 2](#) delineates the fundamentals of the YOLOv8 model and its enhanced variants; [Section 3](#) elucidates the principles of Kalman filtering, Extended Kalman Filtering, and methods of enhancement; [Section 4](#) concentrates on the experimental testing of the model and the analysis of results; [Section 5](#) offers a thorough summary of the entire work.

2. YOLOv8 object detection network and improvements

The YOLO series excels in object identification because of its exceptional balance between speed and accuracy (Redmon et al., 2016).

Ultralytics launched the most recent version, YOLOv8, in January 2023. This method efficiently and precisely identifies targets on diverse mobile devices and is widely utilized across numerous fields, including object identification, tracking, and segmentation. In comparison to other prevalent object detection algorithms, YOLOv8 provides superior speed and enhanced accuracy, attaining state-of-the-art performance across several jobs. YOLOv8 comprises five variants (YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x) designed for jobs of varying scales. This section concentrates on YOLOv8n, presenting DConv3 deformable convolutions and diverse attention processes, introducing the EMAGAM attention mechanism, and enhancing the feature fusion SPPF module.

2.1. YOLOv8 basic architecture

The YOLOv8 design comprises four fundamental components: the Input, the Backbone network, the Neck network, and the Head network, as seen in [Fig. 1](#).

During the input phase, YOLOv8 employs the Mosaic data augmentation technique and modifies hyperparameters according to the model

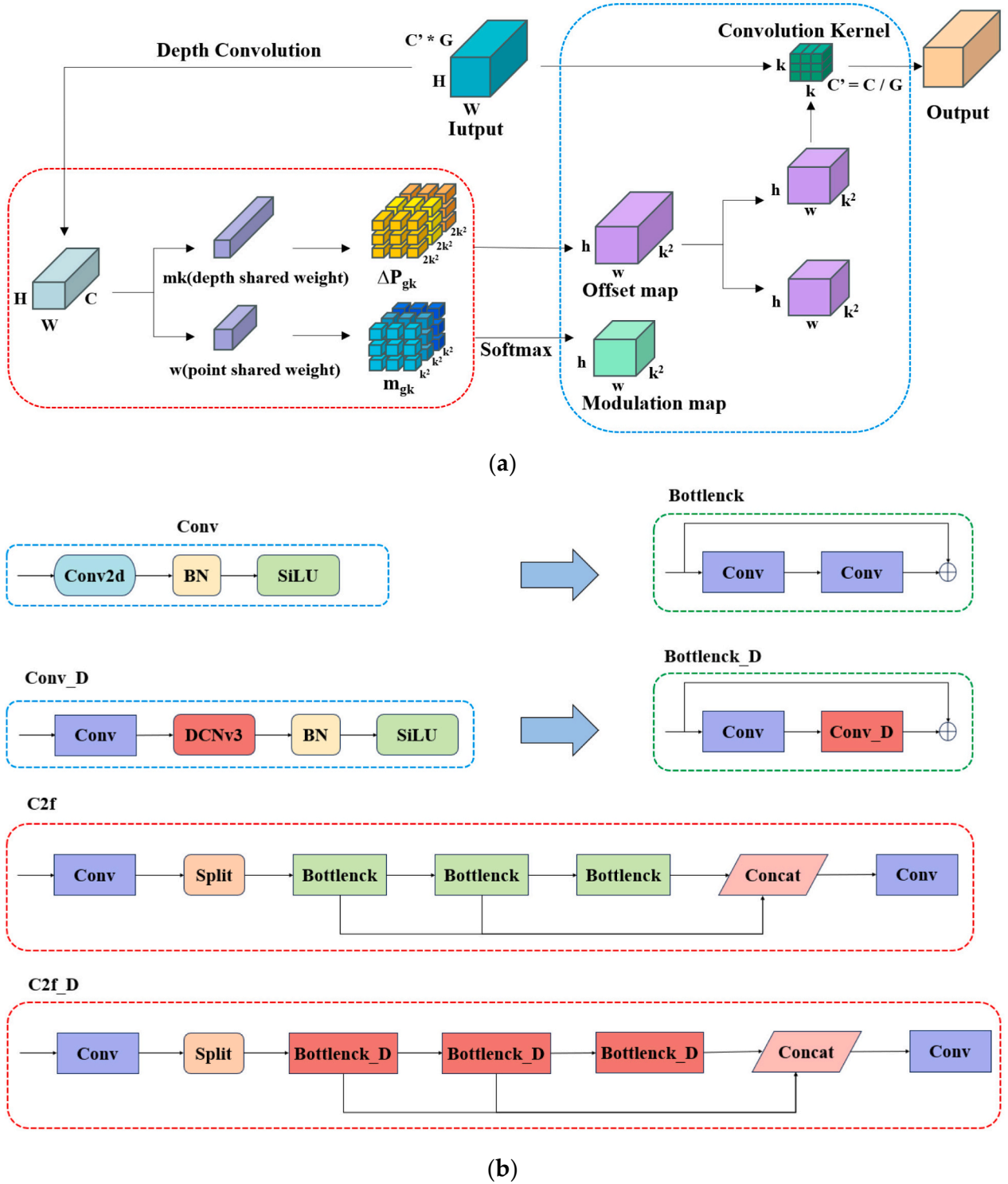


Fig. 2. Module architecture diagram: (a) Architecture of Deformable Convolution DCNv3; (b) Architecture diagram of the combined DCNv3 module.

size, facilitating MixUp and CopyPaste data augmentation for larger models to improve generalization. The backbone network serves as the foundation of the YOLOv8 model, responsible for retrieving critical picture feature information. It presents the distinctive C2F (Cross Stage Partial Network Bottleneck with 2 Convolutions) pre-block to enhance computational efficiency and memory use. In the neck network, YOLOv8 enhances feature extraction from the backbone network by employing the FPNIT + PANISI structure, efficiently merging semantic representation and localization abilities across different scales. In the head

network, YOLOv8 employs a distinct head configuration, effectively partitioning the classification and detection tasks, and incorporates an anchor-free object detection approach to enhance detection speed. In loss computation, YOLOv8 employs a dynamic allocation technique for positive and negative samples. It integrates VFILoss for the computation of classification loss and CIOULoss combined with DFL for the calculation of regression loss.

YOLOv8 exhibits numerous significant enhancements over YOLOv5: Initially, concerning the backbone network, the C2f module supplants

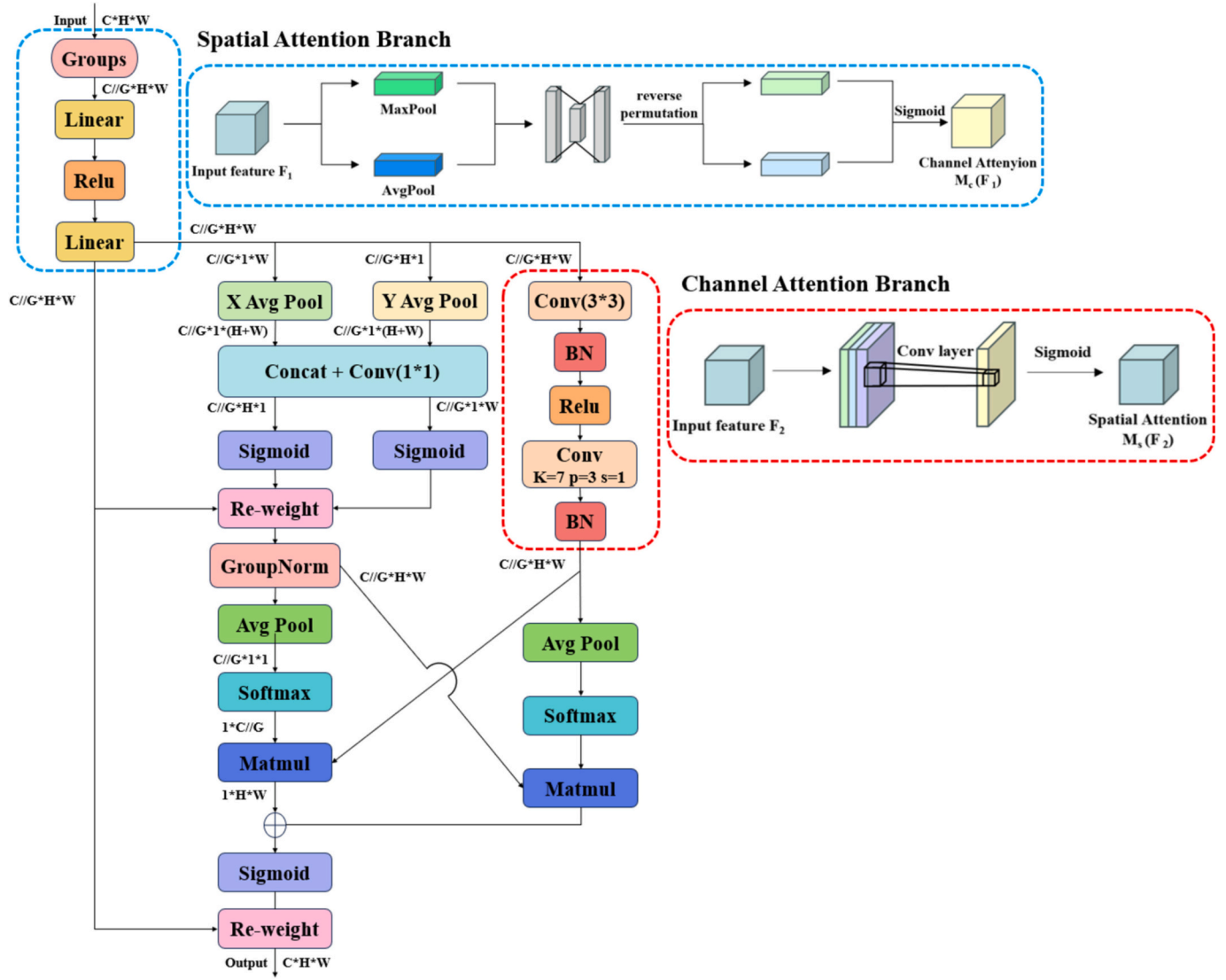


Fig. 3. EMGA Attention Module.

the C3 module, resulting in a more lightweight configuration and improved feature fusion capabilities. Secondly, the neck network replaces the C3 module with the C2f module and removes the convolution module in the PAN-FPN upsampling phase. YOLOv8 employs an Anchor-Free paradigm for loss functions, utilizing BCE Loss for classification and CIOULoss combined with DFL (Distribution Focal Loss) for regression. The sample allocation technique has been revised to employ the Task-AlignedAssigner's dynamic matching of positive and negative samples, replacing the previous IOU matching or unilateral proportion allocation method.

YOLOv8 incorporates exceptional technologies from the realm of real-time object identification into its architecture, preserving and enhancing the core principles of YOLOv5. The model demonstrates robust performance and scalability by integrating the ELAN architecture from YOLOv7, the distinctive C2F module, and the Anchor-Free object detection approach (Wang et al., 2023a).

2.2. Introduction of deformable convolution DCNv3

In contrast to conventional convolution techniques, deformable convolution dynamically modifies the weights of convolutional kernels, effectively tackling the limitation of fixed-size and proportionate rectangular structures that cannot accommodate target geometric

deformations. Conventional convolution techniques frequently neglect essential information on the target during feature extraction, as the receptive field fails to adequately capture the target's geometric deformations, thereby diminishing overall performance. Deformable convolution enhances model performance by modifying the weights of convolutional kernels to better accommodate various image shapes. Moreover, deformable convolution enhances the effective receptive field, facilitating improved capture of long-range dependencies in pictures, and is widely employed in computer vision applications. This new convolution method allows the model to dynamically adjust to animal targets of diverse shapes and sizes, hence improving feature extraction efficiency and accuracy.

This paper presents a deformable convolution, DCNv3, to improve visual feature extraction and enhance model generalization and resilience (Wang et al., 2023b). DCNv3 enhances model performance and efficiency through the implementation of parameter sharing and multi-group techniques. In comparison to earlier iterations such as DCNv2 (Zhu et al., 2019), DCNv3 has enhanced application and representational capacity, rendering it more appropriate for extensive visual tasks. Its advantages encompass enhanced parameter efficiency, accelerated inference speed, an expanded effective receptive field, and augmented representational capacity. The benefits of DCNv3 have resulted in its extensive utilization in computer vision domains, including picture

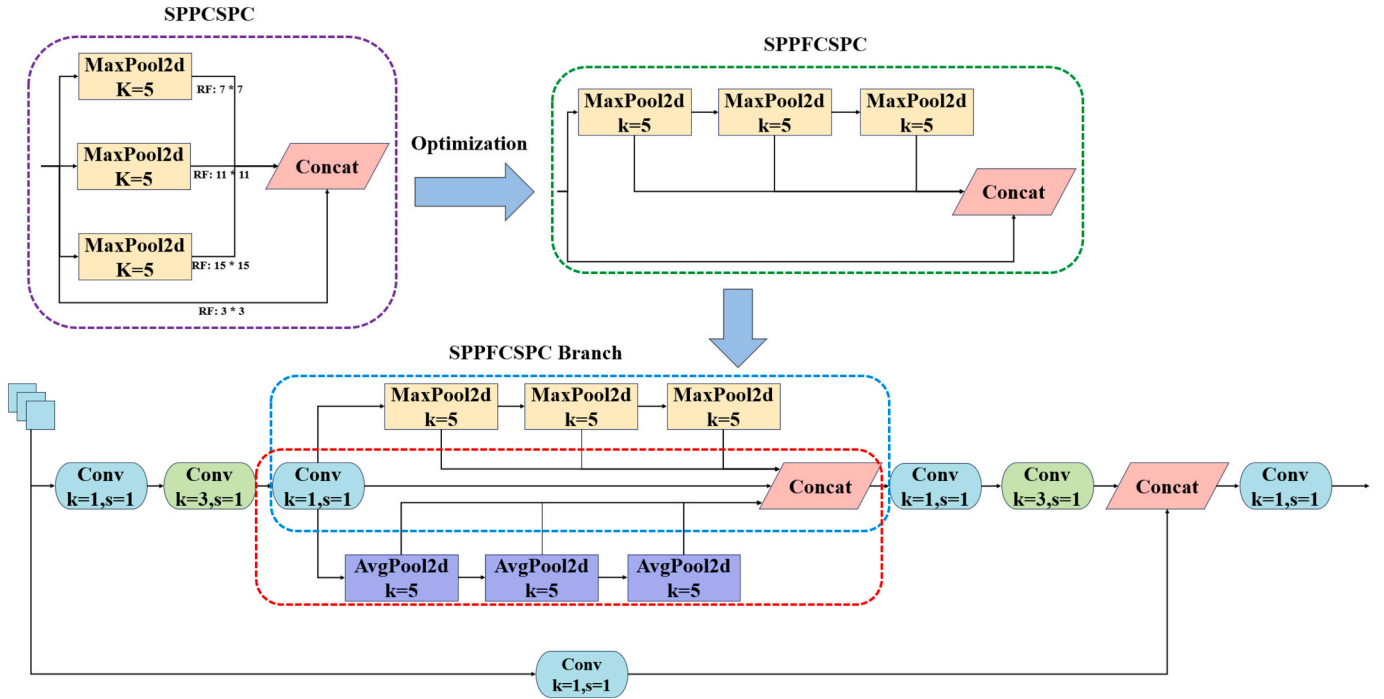


Fig. 4. ASPFC Module.

classification, object identification, and semantic segmentation. DCNv3 can be expressed as Eq. (1).

$$y(p_0) = \sum_{g=1}^G \sum_{k=1}^K w_g m_{gk} x_g(p_0 + p_k + \Delta p_{gk}) \quad (1)$$

In Eq. (1), G represents the overall count of grouped groups. For the g -th group, w_g represents the position-agnostic projection weight of the group, where d_g represents the group's dimension. The offset of the K -th sampling point in the m_{gk} group is standardized across the K dimension by the softmax function. x_g denotes the segmented input feature map, and Δp_{gk} denotes the offset for the sampled grid position p_k in the g -th group. This paper integrates the DCNv3 module into the YOLO network architecture, specifically replacing the existing c2f module, thereby enhancing both the model's performance and efficiency. The module architecture is illustrated in Fig. 2.

2.3. Construction of the EMGA attention module

The EMGA (Efficient Multi-Scale and Global Attraction Mechanism) attention mechanism is a novel module that improves the efficacy of deep learning networks in computer vision applications. The module integrates two distinct attention processes, specifically EMA (Efficient Multi-Scale Attention) (Ouyang et al., 2023) and GAM (Global Attraction system) (Liu et al., 2021), to address their individual shortcomings, resulting in a robust and efficient attention system. The structure is illustrated in Fig. 3, comprising primarily (1) Channel Attention Branch: utilizing partial channel reconfiguration and a grouping design from EMA, it accomplishes channel information preservation and recalibration through operations including group normalization, 1×1 convolution, BatchNorm, and ReLU activation. The Spatial Attention Branch, influenced by the spatial attention mechanism in GAM, captures global context information through layers including AdaptiveAvgPool2d, Conv2d, BatchNorm, and ReLU, thereby augmenting the representational capacity of spatial features. Attention Fusion: The normalization of attention weights from the two branches is performed using the Softmax function, followed by the fusion of channel and spatial attention information by matrix multiplication to derive the final attention weights. Feature Rescaling: EMGA's multi-scale feature fusion

technique, facilitated by EMA, enhances its capacity to gather feature information across several scales, hence improving its proficiency in managing multi-scale data.

Fig. 3 illustrates that in the channel attention branch, EMGA employs EMA's partial channel reconstruction and grouping design, successfully maintaining the integrity of each channel's information through operations including group normalization, 1×1 convolution, BatchNorm, and ReLU, thereby mitigating the negative impacts associated with channel dimension reduction. Simultaneously, by using GAM's global context perception mechanism, EMGA employs layers including AdaptiveAvg-Pool2d, Conv2d, BatchNorm, and ReLU within the spatial attention branch to acquire global context information, thereby augmenting the network's recognition of global characteristics. During the attention fusion phase, EMGA normalizes the channel and spatial attention weights utilizing the Softmax function and amalgamates the information from the two branches via matrix multiplication. Furthermore, EMGA adopts EMA's technique for distributing computation tasks to parallelize channel and spatial attention, thereby significantly decreasing computational expenses and enhancing model efficiency. The multi-scale feature fusion method is utilized, allowing EMGA to more effectively gather feature information across various scales, hence improving its capacity to manage multi-scale data. The distinct advantages are: Channel information preservation: adopting EMA's technique for channel information retention, effectively maintaining the integrity of each channel through grouping design and group normalization, while mitigating negative impacts associated with channel dimensionality reduction. Global context perception: implementing GAM's technique to capture global features via the spatial attention branch, hence augmenting the network's awareness of global context. Reduction in computational costs: by adopting EMA's computation task distribution approach of the attention mechanism, computational expenses are significantly diminished by parallel processing of channel and spatial attention, hence enhancing model efficiency. Multi-scale feature fusion: By employing EMA's multi-scale feature fusion technique, EMGA may more effectively capture feature information across various scales, hence improving its capacity to manage multi-scale information.

In summary, EMGA optimally utilizes the benefits of both EMA and GAM by amalgamating two distinct attention mechanisms, hence

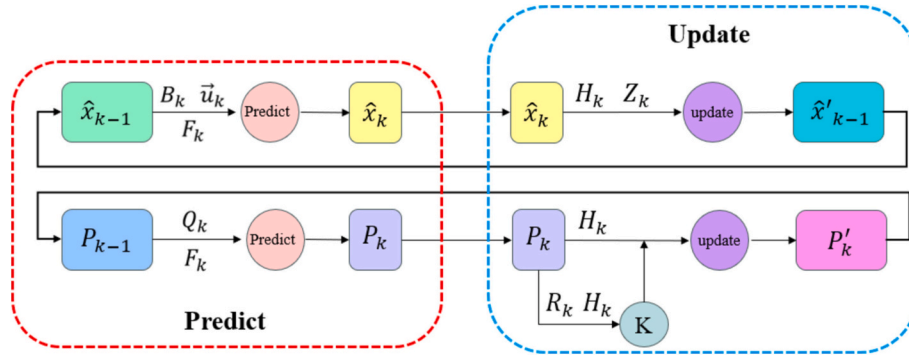


Fig. 5. Kalman Filter Information Flow.

creating a more holistic and efficient attention module. In this work, the EMGA attention mechanism outperforms conventional attention mechanisms and other sophisticated methods. This enhancement in performance arises from the efficient retention of channel information, the incorporation of global contextual awareness, the minimization of computational expenses, and the adaptable implementation of multi-scale feature fusion techniques. Consequently, EMGA offers a more effective attention mechanism for computer vision applications within deep neural networks.

2.4. Construction of the ASPFC multi-feature fusion module

This research introduces a new ASPFC (Average Spatial Pyramid Pooling Concurrent Spatial Pyramid Convolution – Fast) module to mitigate errors arising from size discrepancies across various animals, as illustrated in Fig. 4. This module is a feature fusion component utilized in object identification models, designed to address the constraints of conventional convolutional neural networks in processing input images of varying scales and enhancing feature transmission to augment detection accuracy and efficiency. The ASPFC module refines the SPPCSPC (Spatial Pyramid Pooling Concurrent Spatial Pyramid Convolution) branch into the SPPFCSPC (Spatial Pyramid Pooling Concurrent Spatial Pyramid Convolution - Fast) and incorporates it with an Avg-Pool2d branch, thereby improving the model's ability to perceive wild animals at multiple scales and mitigating local information loss.

The SPPCSPC branch amalgamates two fundamental technologies: Spatial Pyramid Pooling (SPP) (He et al., 2015) and Cross Stage Partial Connections (CSP), employing spatial pyramid pooling to resolve the limitation of classic convolutional neural networks, which can only handle pictures of a predetermined size. The SPPCSPC branch employs maximum pooling procedures at various scales to extract global and local features, while utilizing 1×1 convolution to reduce the number of channels, hence diminishing the model's computing burden and improving performance. This study optimizes the SPPCSPC branch, resulting in the SPPFCSPC module, which enhances speed while preserving the same receptive field. This design minimizes parameters and computing burden, enhancing the model's computational efficiency. Secondly, this work presents an AvgPool2d branch to address the issue of local information loss resulting from maximum pooling operations. The AvgPool2d branch mitigates the loss of detailed characteristics resulting from maximum pooling operations, hence enabling the model to capture more extensive and nuanced information. The incorporation of this average pooling branch enhances the model's generalization capacity and bolsters its robustness in practical applications, enabling effective adaptation to targets of diverse scales. Ultimately, the ASPFC module significantly improves the model's performance by integrating seven distinct feature maps from the SPPCSPC and AvgPool2d branches. This feature fusion strategy enhances the model's sensitivity to features of varying scales. It improves its anti-interference capacity by offering a more comprehensive feature representation, allowing it to manage

complicated scenarios more effectively in real-world applications.

This work effectively mitigates local data loss and enhances model performance by incorporating the average pooling branch. The innovative design adeptly reconciles the requirements of feature dimension reduction and feature fusion, so enhancing the SPPCSPC module and markedly increasing the accuracy and efficiency of detecting operations.

3. Kalman filter improvement

The YOLOv8 model presently incorporates the BoT-SORT multi-object tracking technique, which amalgamates motion and appearance data, mitigates camera motion, and enhances the precision of the Kalman filter state vector (Aharon et al., 2022). This integration markedly enhances performance measures including MOTA, IDF1, and HOTA. It is important to note that the filter employed in the YOLOv8 model is a linear Kalman filter. This work enhances model performance by introducing a nonlinear filter, the Extended Kalman Filter, and implementing enhancements inside the BoT-SORT tracking module of the YOLOv8n model.

3.1. Kalman filter (KF) and extended Kalman Filter (EKF)

The Kalman Filter (KF) is an optimal estimation technique predicted on a state-space model, extensively utilized in signal processing, control systems, navigation systems, and robotics (Chen et al., 2021). This filter assesses the status of a dynamic system based on a series of imprecise and noisy observations, optimizing for minimal mean-square error in state estimate. Target tracking algorithms are generally segmented into two phases: forecasting and updating. During the prediction phase, the existing condition of the target, encompassing location and velocity, is estimated utilizing the system model to obtain the projected value. The expected value is revised and modified throughout the update phase utilizing actual observation results to derive the final state estimation value. (See Fig. 5) The Extended Kalman Filter (EKF) enhances the Kalman Filter to tackle the state estimation challenge in nonlinear system and observation models (Yang and Baum, 2017). It functions as an extension of the Kalman Filter for nonlinear systems by linearizing the nonlinear system using observation models to estimate the system state. The Extended Kalman Filter addresses nonlinearity by linearizing the system and observation models, in contrast to the standard Kalman Filter. Jacobian matrices approximate nonlinear functions during the forecasting and updating phases, converting the nonlinear problem into a linear framework.

3.2. Extended Kalman filter improvement

In multi-target tracking systems, the thresholding of measurement-prediction residuals is a crucial phase, intended to identify measurements that are proximate to the expected position for the purpose of updating the target state. In Kalman filters, Mahalanobis distance is

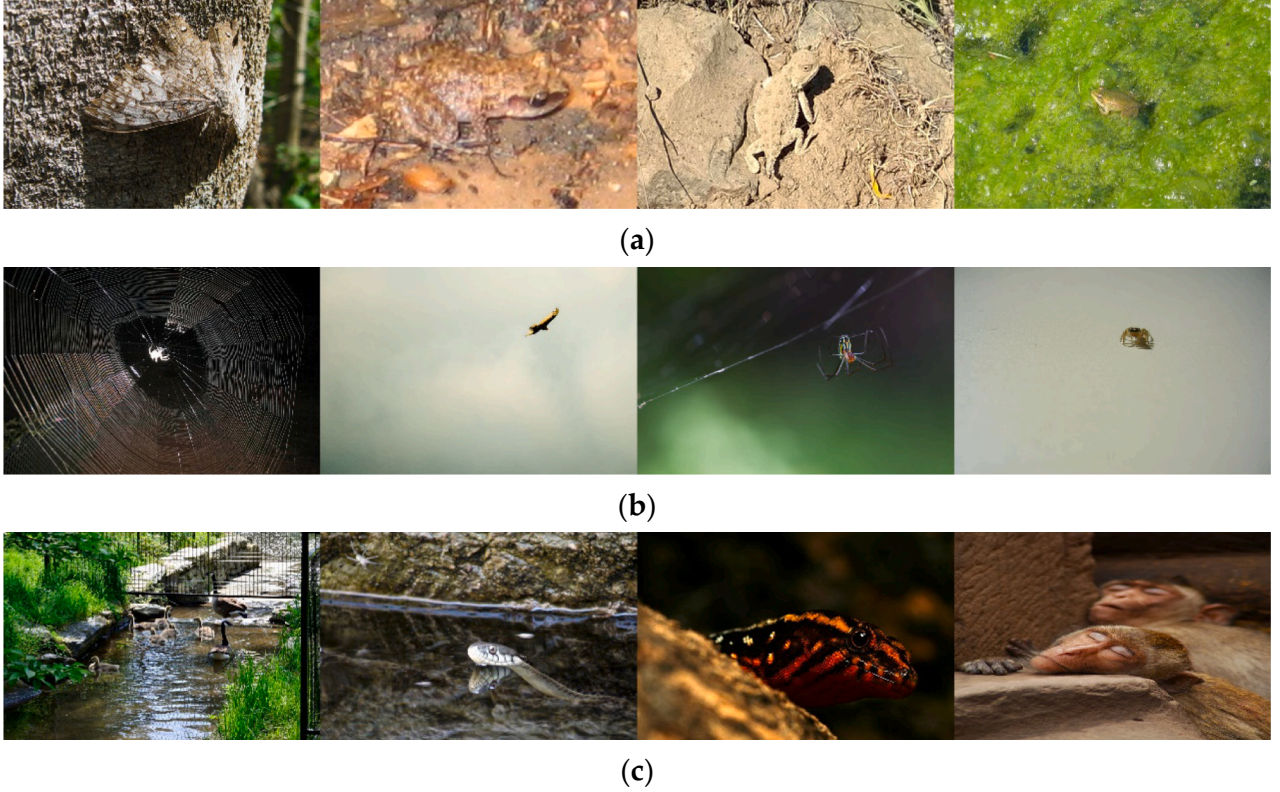


Fig. 6. Examples of different types of datasets: (a) Wildlife in complex backgrounds; (b) Wildlife in small objects; and (c) Wildlife under occlusions.

frequently employed to assess the similarity of measurement-prediction residuals, with a chi-square distribution cutoff to establish the criterion. The enhancement suggested in the study primarily concentrates on the computation of the Mahalanobis distance and the selection of the chi-square distribution threshold.

Initially, the computation of the Mahalanobis distance employed a function that incorporates Eq. (2) to resolve a system of linear equations, potentially leading to performance inefficiencies. The initial formula for computing the Mahalanobis distance is presented in Eq. (2).

$$d^2 = (z - \hat{z})^T S^{-1} (z - \hat{z}) \quad (2)$$

In Eq. (2), z represents the measurement value, $\hat{z}$ represents the predicted value, and S is the covariance matrix. Subsequently, a chi-square distribution is used to set a threshold, meaning that if d^2 is less than a certain threshold, the measurement value is considered acceptable. The improvement in this study mainly focuses on the calculation method of the Mahalanobis distance and the setting of the threshold. The improved approach involves directly calculating the Mahalanobis distance through matrix operations to enhance computational efficiency. By utilizing matrix operations, the calculation steps can be optimized into a single operation without loops or multiple function calls, thereby improving computation speed and efficiency. Specifically, the original method employs a step-by-step solution approach. First, the covariance matrix S is decomposed into a lower triangular matrix L , such that $S = LL^T$. Then, by solving the linear equation $Lz = d$, an intermediate variable z is obtained, where d is the vector of differences between the measurements and the mean. Finally, the squared Mahalanobis distance is obtained by calculating the sum of squares of z , that is, $d^T S^{-1} d = z^T z$. The improved method first calculates the inverse matrix of the covariance matrix S , S^{-1} , and then computes the product of the difference vector d and the inverse covariance matrix S^{-1} through matrix multiplication, that is, $d^T S^{-1} d$.

Secondly, the original method utilized pre-established chi-square distribution thresholds, which were computed beforehand depending on

the dimensions of the Kalman filter state vector. The enhanced method entails the dynamic selection of the chi-square distribution threshold, tailored to the specific application context and data attributes, utilizing the cumulative distribution function of the chi-square distribution. The equation is presented in Eq. (3).

$$F(x; k) = \int_0^x \frac{1}{2^{k/2} \Gamma(k/2)} t^{(k/2)-1} e^{-t/2} dt \quad (3)$$

In Eq. (3), k represents the degrees of freedom (typically the dimension of the covariance matrix), x is the threshold for the Mahalanobis distance and $\Gamma(\cdot)$ denotes the gamma function.

Furthermore, establishing a dynamic threshold requires the specification of a desired confidence level (represented by p). This study subsequently determines the threshold value associated with the confidence level by computing the inverse function (percentile) of the chi-square distribution.

$$\text{Threshold} = \chi_k^{-1}(p) \quad (4)$$

In Eq. (4), $\chi_k^{-1}(p)$ represents the p -th percentile of the chi-square distribution, where k is the degrees of freedom. The advantage of this improvement is that it can better adapt to different data distributions and tracking environments, thereby enhancing the accuracy and robustness of the tracking system.

In conclusion, these enhancements enable multi-target tracking systems to more effectively adjust to various inputs and scenarios, hence improving real-time performance and tracking precision. Enhancing the measurement update process is essential for increasing the MOTA (Multiple Object Tracking Accuracy) and HOTA (Highly Overlapped Target Accuracy) metrics, since it directly influences the accurate identification and tracking of targets. Consequently, these improvements are crucial for the quality and credibility of the article.

4. Experiment and discussion

4.1. Dataset construction and experimental environment configuration

This experiment was conducted on an Ubuntu 22.04 operating system featuring two A800 (80G) Graphics Processing Units (GPUs). The software stack included PyTorch 2.1.1, torchvision 0.16.1, torchaudio 2.1.1, CUDA 12.1, and Python 3.9. The experiment was conducted across 200 epochs, utilizing a batch size of 256, an initial learning rate of 0.001, and a confidence threshold of 0.5. Model optimization was conducted via the Adam optimization technique.

The collection comprises 6603 photos representing 10 distinct kinds of wild animals, including butterflies, frogs, lizards, snakes, and monkeys, together with a 4 K quality, 60fps movie. The data were acquired from publically accessible web sources, and the majority of the images and videos feature intricate backgrounds (such as mud, grass, rivers, trees, etc.) and include little targets. (See Fig. 6) Following the use of Diffusion data augmentation, the total number of photos rose to 17,169, with each image annotated utilizing the LabelImg tool. The dataset was ultimately partitioned into training, validation, and test sets in an 8:1:1 ratio.

4.2. Experimental evaluation metrics

This work employed the following measures for performance testing and assessment in the object detection phase: precision (Tharwat, 2020), recall (Yacouby and Axman, 2020), mean Average Precision (mAP) (Hossin and Sulaiman, 2015; Padilla et al., 2020), and Floating Point Operations (FLOPs) (Hunger, 2005). The mAP was computed at IoU levels of 0.50 and 0.50–0.95, referred to as mAP50 and mAP50–95, respectively. During the object tracking phase, this study utilized High-Order Tracking Accuracy (HOTA) (Luiten et al., 2021), Multiple Object Tracking Accuracy (MOTA) (Bernardin and Stiefelhagen, 2008), and ID F1 measures (Chicco and Jurman, 2020).

1. Precision

Precision refers to the ratio of correctly predicted positive samples by the model. In object detection tasks, precision assesses the overlap between the model's anticipated and actual bounding boxes. A prediction is considered accurate if the model's predicted bounding box overlaps with the actual bounding box. It is delineated in Eq. (5), where TP denotes True Positives and FP signifies False Positives.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (5)$$

2. Recall

Recall denotes the proportion of true positive samples accurately recognized by the model. In object detection tasks, recall evaluates the model's ability to identify all accurate bounding boxes. A sample is considered accurately recalled when there is an overlap between the actual and projected bounding boxes. It is delineated in Eq. (6), where TP signifies True Positives and FN denotes False Negatives.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (6)$$

3. Mean average precision (mAP)

mAP (Mean Average Precision) denotes the average accuracy across all categories within the complete dataset. It demonstrates the model's comprehensive performance across multiple categories. mAP50 refers to the mean Average Precision value at a 50 % Intersection over the Union (IoU) threshold. In contrast, mAP50–95 is a more rigorous metric that

calculates the mAP values within the IoU threshold range of 50 % to 95 % and averages them. This statistic may more precisely evaluate the model's performance across various IoU thresholds, offering a more thorough performance assessment. It is delineated in Eqs. (8), (9), and (10).

Accuracy (AP) represents the area under the Precision-Recall curve, as specified in Eq. (7), where precision(r) is the precision at recall rate r .

$$\text{AP} = \int_0^1 \text{precision}(r) dr \quad (7)$$

In computing mAP (Mean Average Precision), the area under the Precision-Recall curve is computed for each class, defined as in Eq. (8), where n is the number of classes and AP_i is the Average Precision for the i -th class.

$$\text{mAP} = \frac{1}{n} \sum_{i=1}^n \text{AP}_i \quad (8)$$

mAP50 represents the mAP value at a 50 % Intersection over the Union (IoU) threshold, and its calculation formula is defined as in Eq. (9), where AP_i^{50} is the precision for the i -th class at a 50 % IoU threshold.

$$\text{mAP50} = \frac{1}{n} \sum_{i=1}^n \text{AP}_i^{50} \quad (9)$$

mAP50–95 is the average of mAP values within the IoU threshold range from 50 % to 95 %, and its calculation formula is as given in Eq. (10), where AP_i^{50-95} is the precision for the i th class within the IoU threshold range from 50 % to 95 %.

$$\text{mAP50} - 95 = \frac{1}{n} \sum_{i=1}^n \text{AP}_i^{50-95} \quad (10)$$

4. Floating point operations (FLOPs)

Floating Point Operations (FLOPs) characterize a model's computational complexity and are frequently employed as an indirect metric to evaluate the speed of neural network models. It denotes the number of floating-point operations required to process an image and facilitates the comparison of inference speeds across different techniques. FLOPs directly indicate the computational burden and operational efficacy of the model, hence playing a crucial role in assessing model efficiency and optimization initiatives.

5. HOTA (highly overlap tile assignment)

HOTA is a high-order tracking accuracy metric utilized to comprehensively evaluate object recognition and tracking ability. It evaluates the overlap and correlation among targets by framing target assignment as a graph-matching problem and employing the Hungarian algorithm to optimize matching scores. HOTA assesses tracking accuracy by evaluating the overlap between targets and their temporal-spatial consistency. It is defined by the ratio of correctly matched targets to the total targets, as seen in Eq. (11).

$$\text{HOTA} = \frac{\text{Correct Matches}}{\text{Total Targets}} \quad (11)$$

6. MOTA (multiple object tracking accuracy)

MOTA is employed to evaluate the precision of multi-object tracking. It comprehensively assesses components, including false positives, false negatives, discrepancies, and accurate matches. The MOTA metric evaluates tracking accuracy by analyzing the correspondence between actual trajectories and tracking outcomes, specifically by calculating the false positive rate, false negative rate, and inaccurate match rate. It is defined as one minus the aggregate of the false positive rate, false negative rate, and wrong match rate, as illustrated in Eq. (12).

Table 1
Comparison of Evaluation Metrics Before and After Data Augmentation.

Model	precision	recall	mAP50	mAP50-95	FLOPs
YOLOv8n	84.70 %	72.10 %	77.68 %	56.05 %	8.2G
YOLOv8n*	88.35 %	80.83 %	83.97 %	63.28 %	8.2G

$$MOTA = 1 - (\text{False Positive} + \text{False Negative} + \text{Mismatch}) \quad (12)$$

7. IDF1 (identity F1 score)

IDF1 evaluates the precision and recall of identification recognition. It accounts for identity alterations throughout the tracking process and denotes the harmonic mean of ID precision and ID recall. IDF1 provides a more thorough assessment of the tracking system's ID recognition efficacy. The formula is presented in Eq. (13).

$$IDF1 = \frac{2 \cdot \text{ID Precision} \cdot \text{ID Recall}}{\text{ID Precision} + \text{ID Recall}} \quad (13)$$

4.3. Evaluation and analysis of improved YOLOv8 model

4.3.1. Analysis of the effectiveness of diffusion data augmentation

Data augmentation is a prevalent strategy in data science employed to enlarge the dataset by incorporating slightly altered current data or synthetically generating new data. Adequate data is essential for object detection activities. Consequently, the Stable Diffusion model was utilized for data augmentation in this experiment. Stable Diffusion accomplishes picture improvement and image-to-image transition via stable diffusion techniques. This model enhances image resolution and incorporates finer details or noise. Moreover, it modifies and augments the picture data, yielding a comprehensive and varied dataset for training and research purposes. In this experiment, each image was sampled twice for data augmentation with a noise intensity of 0.4. Initially, a variational autoencoder (VAE) was utilized to compress the images from pixel space to latent space, thereby encapsulating the fundamental semantic contents of the images. Subsequently, in the forward diffusion process, iterative steps methodically introduced Gaussian noise to the compressed latent representations. Subsequently, by employing the cross-attention mechanism within the UNet model, the

Diffusion Models were converted into more adaptable conditional image generators. The encoder was ultimately implemented to facilitate regulated production of image-to-image operations. The low-dimensional space vector was subsequently upsampled and enhanced to produce entire images via the image decoder.

This study independently trained the YOLOv8n module on datasets prior to and after the implementation of data augmentation to assess its efficacy. To rigorously assess the model's efficacy, all parameters, aside from the varying datasets, were maintained in accordance with those previously specified. The model without data augmentation is designated as YOLOv8n, while the model with data augmentation is indicated as YOLOv8n* with an asterisk in the top right corner.

Table 1 and Fig. 7 demonstrate that the model, following data augmentation (YOLOv8n*), shows substantial enhancements across all assessment measures. Precision rises from 84.70 % to 88.35 %, recall ascends from 72.10 % to 80.83 %, mAP50 advances from 77.68 % to 83.97 %, and mAP50-95 improves from 56.05 % to 63.28 %. The results indicate that employing the Stable Diffusion model for data augmentation offers significant benefits in improving the effectiveness of object detection models. Upon comparing the FLOPs (Floating Point Operations) of the two models, it is evident that data augmentation does not elevate the computational load; both models sustain FLOPs at 8.2G. It demonstrates that data augmentation enhances model performance without incurring additional computing expenses, hence increasing its practicality for real-world applications.

In conclusion, utilizing Stable Diffusion data augmentation in object detection tasks is effective. It can improve model performance by augmenting the diversity and complexity of the dataset without imposing extra computational burden. It provides substantial support and direction for research and implementation in object detection.

4.3.2. Effectiveness analysis of DCNv3

This study evaluates the effectiveness of the DCNv3 deformable convolution by comparing the YOLOv8n model with the YOLOv8n-D model, which incorporates the DCNv3 module. Both models are trained on the expanded dataset, maintaining consistency in all parameters except for the model as previously specified. The experimental results are displayed in Table 2.

The model's performance is clearly improved by the use of the modified version of DCNv3 deformable convolution. In comparison to

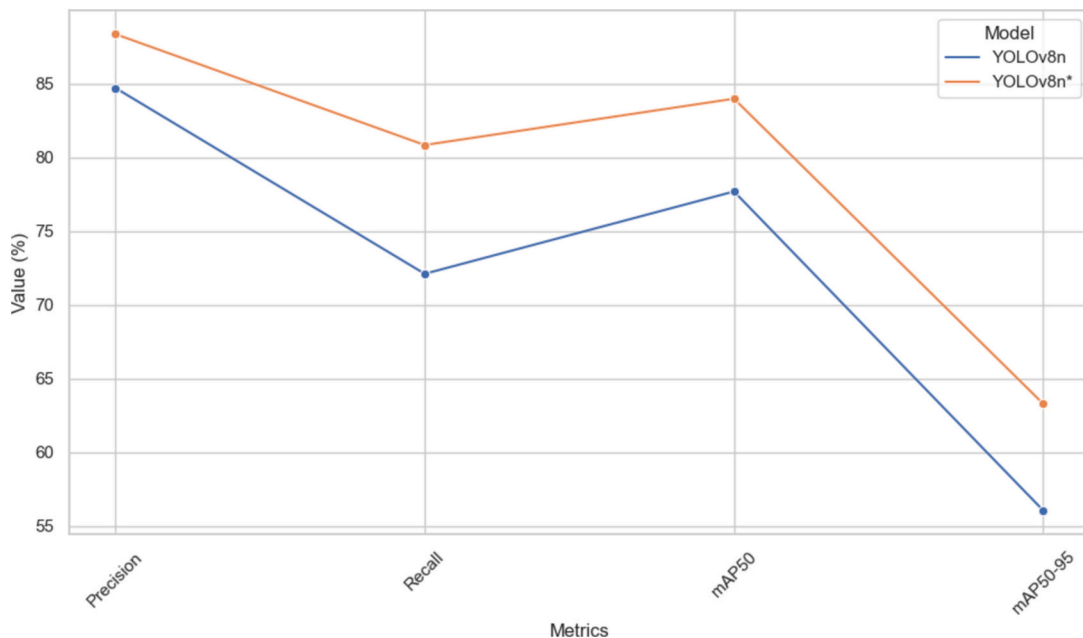


Fig. 7. Performance Comparison between YOLOv8n and YOLOv8n*.

Table 2
Comparison of Evaluation Metrics Before and After Embedding DCNv3 Module.

Model	precision	recall	mAP50	mAP50–95	FLOPs
YOLOv8n*	88.35 %	80.83 %	83.97 %	63.28 %	8.2G
YOLOv8n_DCNv3*	88.76 %	84.87 %	87.06 %	71.20 %	7.9G

the model lacking DCNv3, YOLOv8n_DCNv3* markedly enhances all evaluation criteria. Precision rises from 88.35 % to 88.76 %, recall ascends from 80.83 % to 84.87 %, mAP50 advances from 83.97 % to 87.06 %, and mAP50–95 improves from 63.28 % to 71.20 %. Alongside enhancing model performance, the introduction of DCNv3 also contributes to a degree of improved computing efficiency. The FLOPs of the YOLOv8n_DCNv3* model diminish from 8.2G to 7.9G, signifying that the model with DCNv3 preserves performance advantages while reducing computing demands, thereby improving the model's efficiency. These findings demonstrate that the integration of DCNv3 significantly improves the model's detection accuracy and generalization capability, while also enhancing computational efficiency.

4.3.3. Effectiveness analysis of EMGA attention mechanism

The effectiveness of the proposed EMGA attention module is assessed in comparison to traditional attention modules, including SE, ECA, CBAM, GAM, and EMA. Six attention mechanisms are integrated into the YOLOv8n algorithm architecture, maintaining consistency in all parameters except for the attention module, as previously outlined, with the dataset being the augmented dataset. Fig. 8 displays the experimental evaluation metrics.

Various attention mechanism modules demonstrate distinct impacts on the performance of the YOLOv8n algorithm framework across multiple evaluation metrics: regarding Precision, YOLOv8n_SE* and YOLOv8n_EMA* outperform other models marginally, whilst YOLOv8n_CBAM* falls somewhat short. The accuracy of the EMGA module is comparable to that of various models. Regarding Recall, YOLOv8n_EMGA* demonstrates superior performance, markedly exceeding that of other models, suggesting that the EMGA module excels in target acquisition. Regarding mAP@50 and mAP@50–95,

YOLOv8n_EMGA* markedly surpasses other models, demonstrating that this module enhances the accuracy and stability of detection outcomes. Concerning FLOPs (Floating Point Operations), the computational expenses of the majority of models are comparable; however, the computational cost of YOLOv8n_GAM* is marginally elevated, perhaps resulting in some performance trade-offs. The EMGA module excels in various evaluation criteria, particularly in Recall and mAP. The EMGA module may more effectively capture target features, hence enhancing detection performance.

4.3.4. Effectiveness analysis of ASPFC

This study validates the effectiveness of the suggested ASPFC module by comparing it with other modules, including SPPF from YOLOv5, SPPCSPC from YOLOv7, and the optimized SPPCSPC to SPPFCSPC, alongside the highlighted ASPFC module presented above. Assuming all parameters, except for this module, remain constant as previously discussed, these four modules are integrated into the YOLOv8n model, while the dataset continues to be the augmented dataset.

Table 3 demonstrates that the ASPFC module outperforms YOLOv5's SPPF, YOLOv7's SPPCSPC, and the optimized SPPFCSPC module across many assessment measures. The ASPFC module demonstrates somewhat superior precision compared to other modules and considerably excels in recall, highlighting its effectiveness in target acquisition. This module exhibits superior performance in mAP@50 and mAP@50–95, signifying its capacity to improve the accuracy and consistency of detection results. The computational overhead of the ASPFC module is slightly greater than that of other modules; nevertheless, given the performance

Table 3
Comparison of Evaluation Metrics Before and After Embedding Multi-Scale Feature Fusion Modules.

Model	precision	recall	mAP50	mAP50–95	FLOPs
YOLOv8n* (SPPF)	88.35 %	80.83 %	83.97 %	63.28 %	8.2G
YOLOv8n_SPPCSPC*	88.32 %	79.88 %	84.04 %	62.78 %	9.5G
YOLOv8n_SPPFCSPC*	88.40 %	79.83 %	83.59 %	62.32 %	9.5G
YOLOv8n_ASPFC*	88.99 %	82.14 %	86.86 %	69.49 %	9.7G

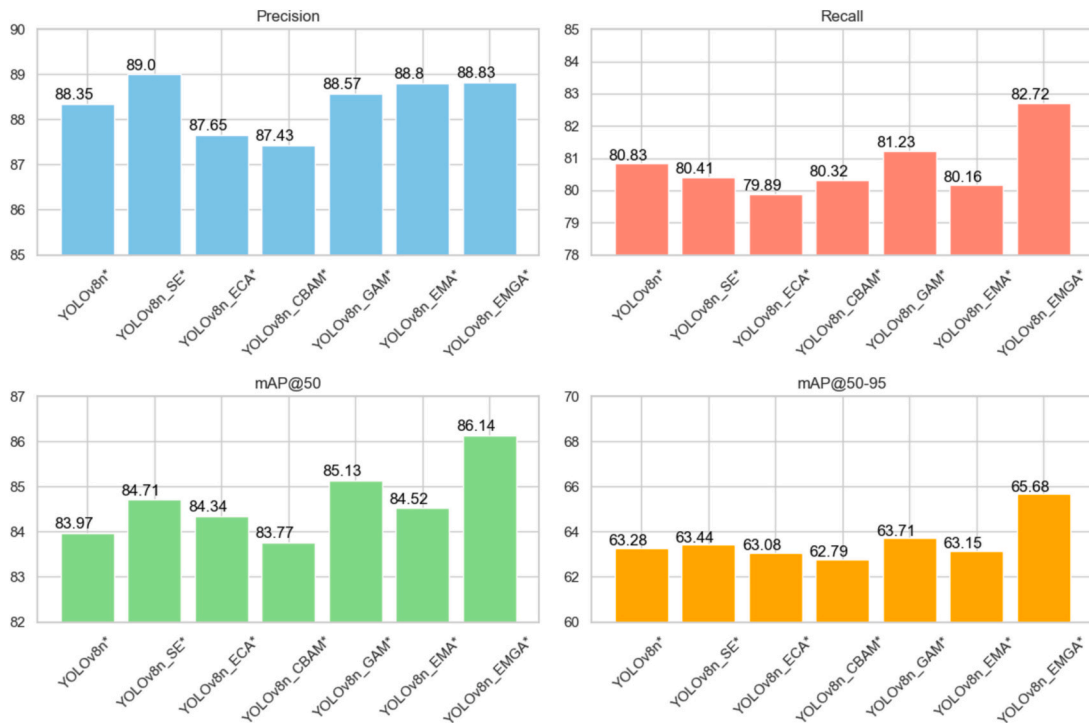


Fig. 8. Comparison of performance of various attention mechanisms.

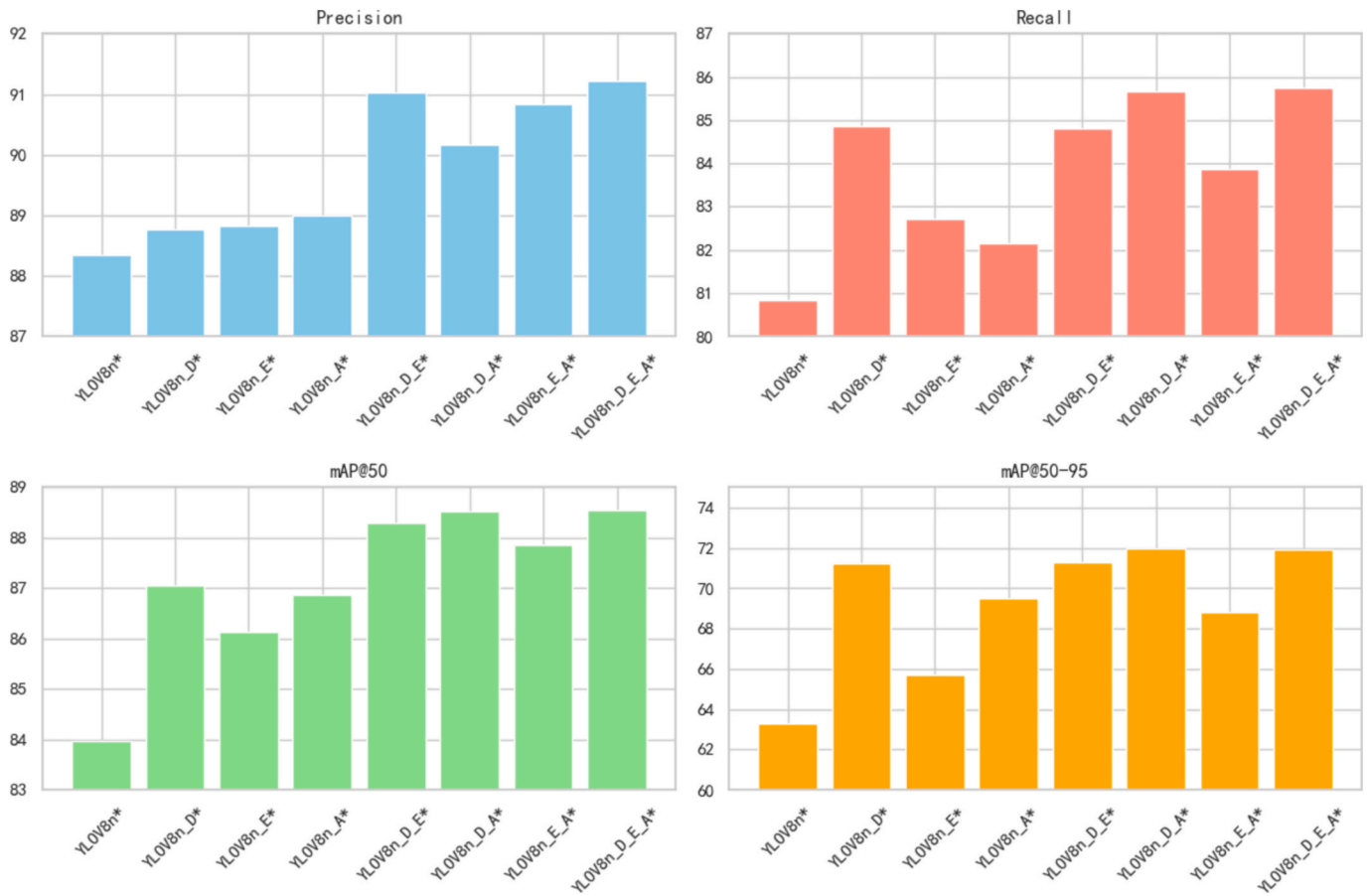


Fig. 9. Analysis of ablation performance.

Table 4
Analysis of ablation experiment results.

YOLOv8n*	DCNv3	EMGA	ASPFC	precision	recall	mAP50	mAP50-95	FLOPs
✓	–	–	–	88.35 %	80.83 %	83.97 %	63.28 %	8.2G
✓	✓	–	–	88.76 %	84.87 %	87.06 %	71.20 %	7.9G
✓	–	✓	–	88.83 %	82.72 %	86.14 %	65.68 %	8.3G
✓	–	–	✓	88.99 %	82.14 %	86.86 %	69.49 %	9.7G
✓	✓	✓	–	91.02 %	84.80 %	88.30 %	71.30 %	8.0G
✓	✓	–	✓	90.17 %	85.66 %	88.52 %	71.95 %	9.4G
✓	–	✓	✓	90.84 %	83.87 %	87.85 %	68.79 %	9.8G
✓	✓	✓	✓	91.22 %	85.75 %	88.54 %	71.94 %	9.5G

enhancement, this increase remains within an acceptable range. The experimental results demonstrate that the ASPFC module improves the performance of the YOLOv8n model, particularly in terms of recall and detection accuracy in object identification tasks.

4.3.5. Results and analysis of ablation experiments

All ablation experiments were conducted on the dataset following data augmentation. This experiment evaluated the impact of the suggested enhancement module on inference performance, keeping all parameters consistent with those previously described and employing the YOLOv8n model as the baseline. Ablation studies were performed on the DCNv3 deformable convolution, EMGA attention mechanism, and ASPFC module.

The outcomes of the ablation tests clearly illustrate the substantial influence of the DCNv3 deformable convolution, EMGA attention mechanism, and ASPFC module on the inference performance of the YOLOv8n model. (See Fig. 9 and Table 4) Specifically: (1) The introduction of the DCNv3 deformable convolution module enhanced the

model's performance across all criteria. Precision rose by 0.41 percentage points, recall rose by 4.04 percentage points, mAP@50 rose by 3.09 percentage points, mAP@50-95 rose by 7.92 percentage points, however FLOPs diminished by 0.3G. The DCNv3 module enhances detection accuracy and diminishes model complexity. The incorporation of the EMGA attention mechanism resulted in enhanced performance across various parameters. Precision rose by 0.48 percentage points, recall rose by 1.89 percentage points, mAP@50 rose by 2.17 percentage points, and mAP@50-95 rose by 2.4 percentage points, but FLOPs remained relatively unchanged. The EMGA module significantly improves the model's proficiency in target capture and detection accuracy. The introduction of the ASPFC module markedly enhanced the model's performance. Precision rose by 0.64 percentage points, recall rose by 1.31 percentage points, mAP@50 rose by 2.89 percentage points, and mAP@50-95 rose by 6.21 percentage points, while FLOPs climbed by 1.5G. It signifies the significant impact of the ASPFC module on improving the model's accuracy.

Moreover, various enhancement modules differentially impact the

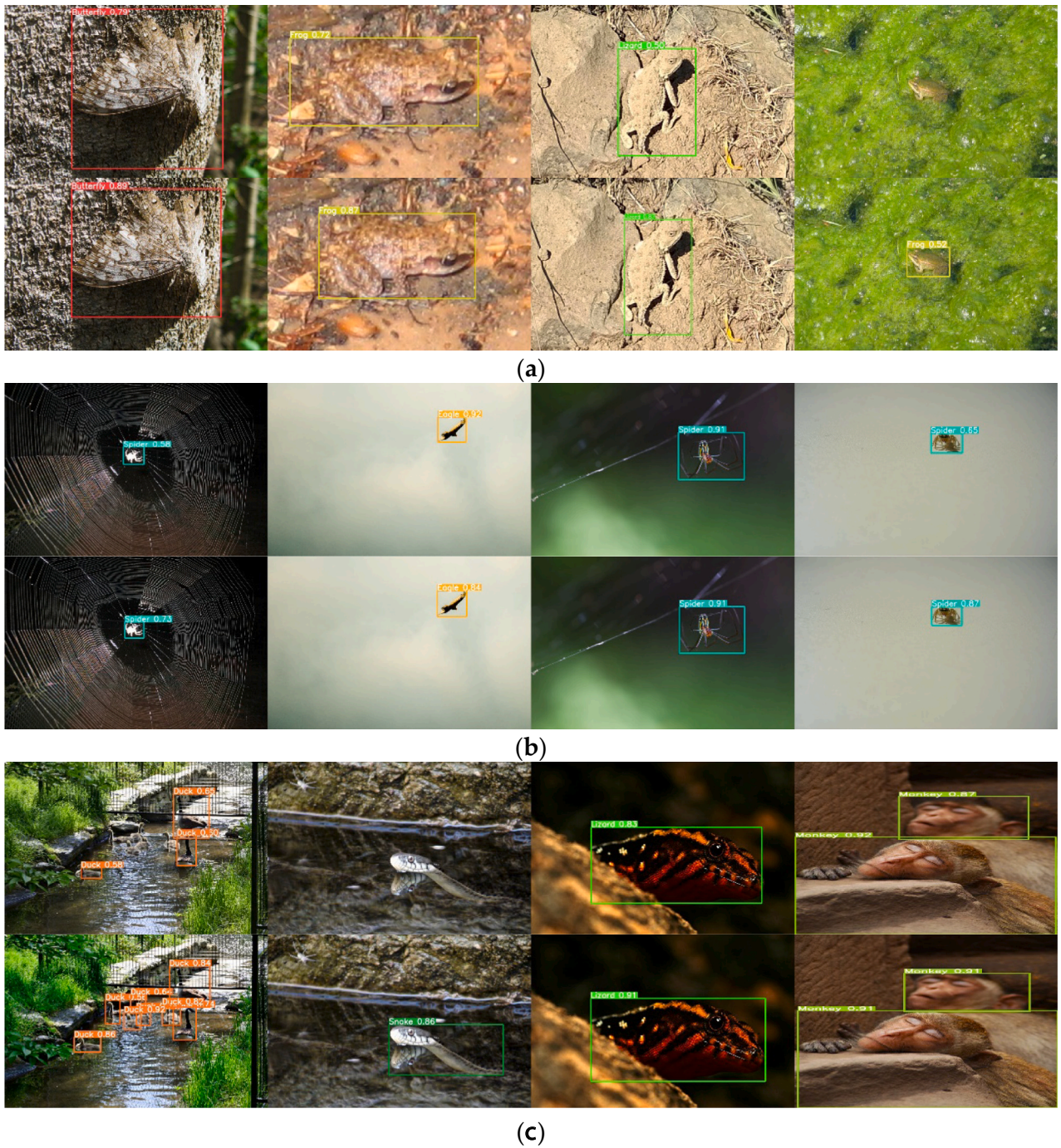


Fig. 10. YOLOv8n(first row) and wilDT-YOLOv8n(second row) detection instances of different types of datasets: (a) The comparison graphs of the detection results of wild animals in complex backgrounds; (b) The comparison graphs of the detection results of wild animals in small targets; (c) The comparison graphs of the detection results of wild animals under occlusion.

performance augmentation of the YOLOv8n model, and the integration of diverse modules can more effectively exploit their own strengths. The model achieves best performance when integrating the DCNv3, EMGA, and ASPFC modules. Precision rose by 2.87 percentage points, recall rose by 4.92 percentage points, mAP@50 rose by 4.57 percentage points, and mAP@50–95 rose by 8.66 percentage points, while FLOPs climbed by 1.3G.

Fig. 10 juxtaposes the outcomes of the YOLOv8n and wilDT-YOLOv8n models in detecting tasks. Upon examining the photographs in group (a) with intricate backgrounds, the enhanced models demonstrated notable advancements in inference accuracy, achieving ranges of

79 % to 89 %, 72 % to 87 %, and 50 % to 87 %, respectively. They successfully identified animal things that prior models had overlooked. In group (b), which concentrates on small targets, there was a marginal decline in accuracy for certain detection tasks; however, the models exhibited considerable stability, with accuracy levels between 58 % and 73 %, 92 % and 84 %, 91 % and 91 %, and 65 % and 87 %. In group (c), comprising images with occlusions, the enhanced models not only detected several targets overlooked by earlier models but also demonstrated overall advancements in detection accuracy.

The ablation tests demonstrate that the DCNv3 deformable convolution, EMGA attention mechanism, and ASPFC module significantly

Table 5
Comparison of Evaluation Metrics Before and After Replacing the Filter.

Filter	HOTA	MOTA	IDF1
Kalman	39.614 %	36.427 %	47.108 %
Extended Kalman	42.236 %	38.775 %	51.282 %

affect the inference performance of the YOLOv8n model. The integration of these enhancement modules can significantly improve the model's performance and stability, providing strong support for the further optimization of object detection tasks.

4.4. Improved extended Kalman filter testing evaluation and analysis

4.4.1. Analysis of the effectiveness of introducing extended Kalman filter

To assess the efficacy of replacing the Kalman filter with the Extended Kalman filter in the YOLOv8 model, the models are tested and evaluated prior to and after the filter substitution, resulting in Table 5 and Fig. 11. All settings remain consistent except for the filter.

As shown in Fig. 12, it can be observed that after replacing the Kalman filter with the Extended Kalman filter, the performance of the model has significantly improved. Especially in terms of HOTA and IDF1 metrics, the application of the Extended Kalman filter has led to an increase of approximately 2–3 percentage points. It is indicated that the Extended Kalman filter performs better in multi-object tracking applications, as it can more accurately estimate the system state, thereby

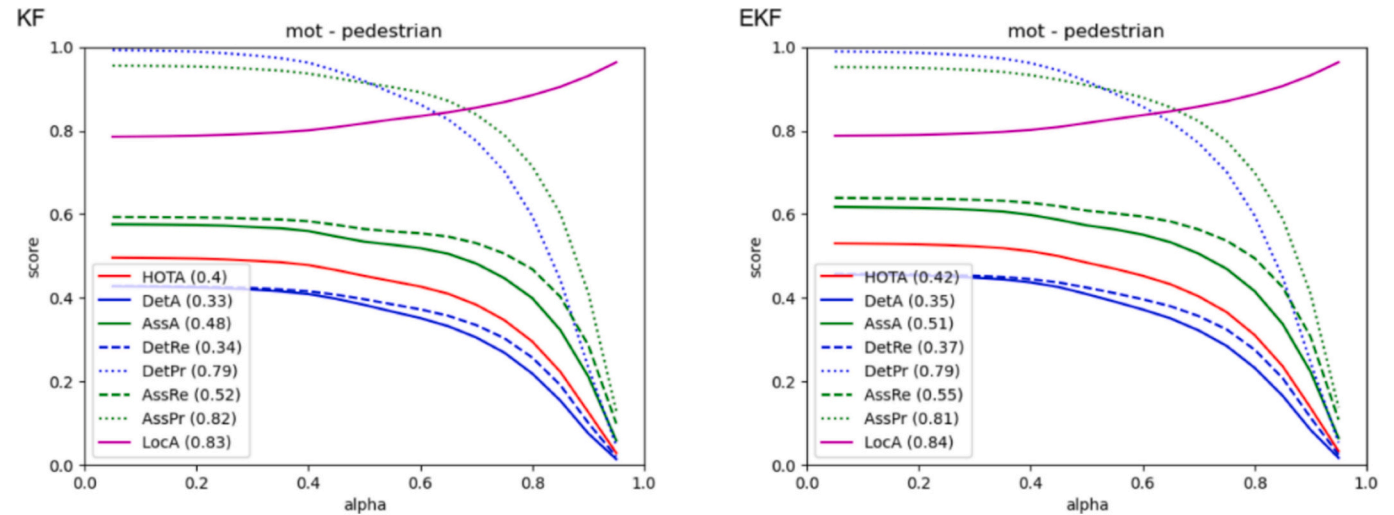


Fig. 11. Pedestrian plot.

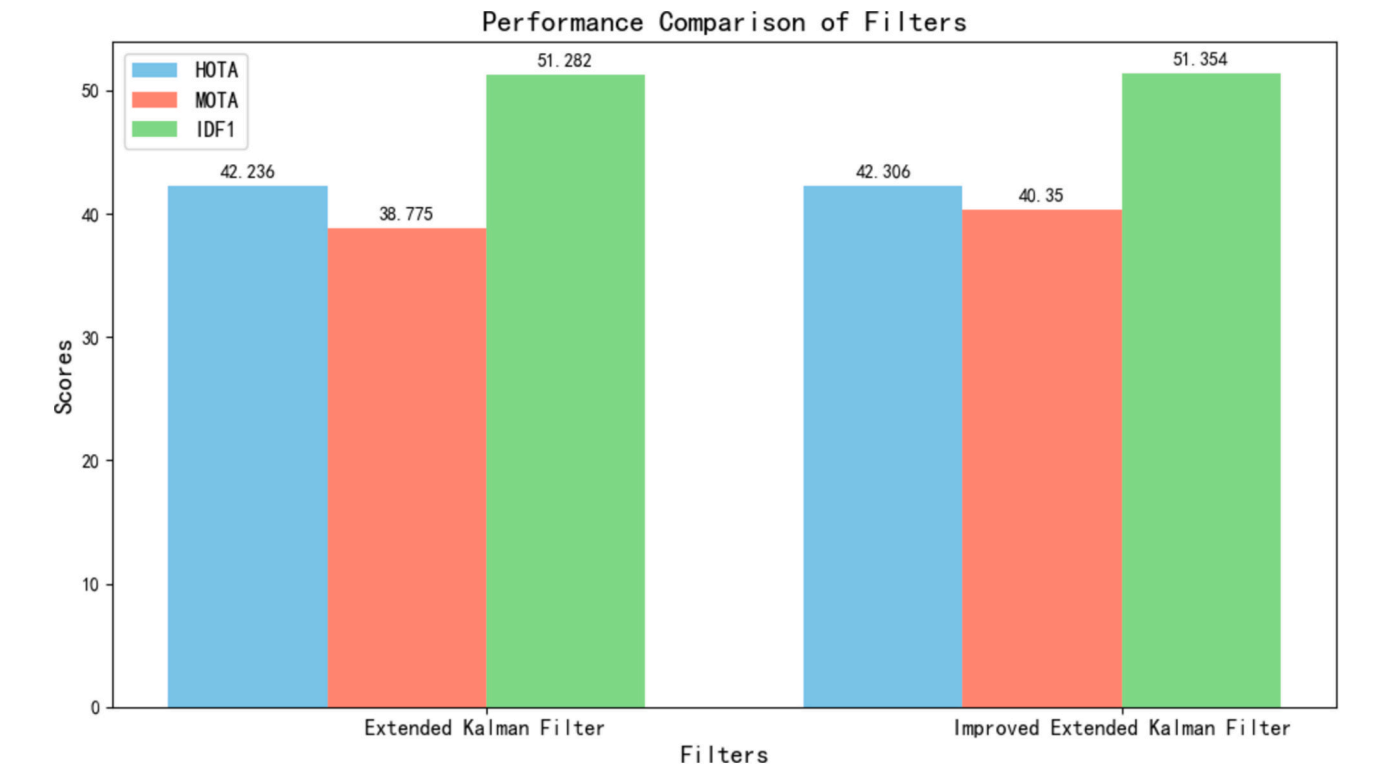


Fig. 12. Performance Comparison of Filters.



Fig. 13. Comparison of Tracking Results Before and After Introducing Extended Kalman Filter.

Table 6
Comparison of Evaluation Metrics Before and After Filter Replacement.

Filter	HOTA	MOTA	IDF1
Original EKF	42.236 %	38.775 %	51.282 %
Improved EKF	42.306 %	40.35 %	51.354 %

enhancing the model’s overall performance. The actual tracking results are depicted in Fig. 13.

Fig. 12 illustrates that the substitution of the Kalman filter with the Extended Kalman filter has markedly enhanced the model’s performance. The implementation of the Extended Kalman filter has resulted in an enhancement of around 2–3 percentage points, particularly with the HOTA and IDF1 measures. The Extended Kalman filter has shown to excel in multi-object tracking applications due to its superior capability in estimating the system state, therefore improving the model’s overall performance. The actual tracking outcomes are illustrated in Fig. 13.

4.4.2. Improved Extended Kalman Filter effectiveness analysis

This study evaluated the performance of the enhanced Extended

Kalman Filter (EKF) by conducting tests on both the improved and original versions, maintaining consistency in all parameters except for the filters. The experimental results are displayed in Table 6 and Fig. 14.

Figs. 15 and 16 demonstrate that the implementation of the enhanced EKF led to an improvement in the model’s performance. The 1.5 % increase in MOTA is particularly significant, suggesting that the proposed enhancement method has improved the performance of the multi-object tracking system to a degree. While the enhancements in other measures, such as HOTA and IDF1, are not markedly significant, they do demonstrate a modest upward trajectory, suggesting that the refined EKF possesses advantages across many evaluation criteria. It further validates the effectiveness of the proposed improvement strategy and establishes a robust foundation for the continued optimization of multi-object tracking systems. Fig. 16 illustrates the actual tracking results.

This work offers significant insights and practical experiences for the selection and parameter adjustment of filters in multi-object tracking systems, presenting viable solutions for real-world target tracking challenges.

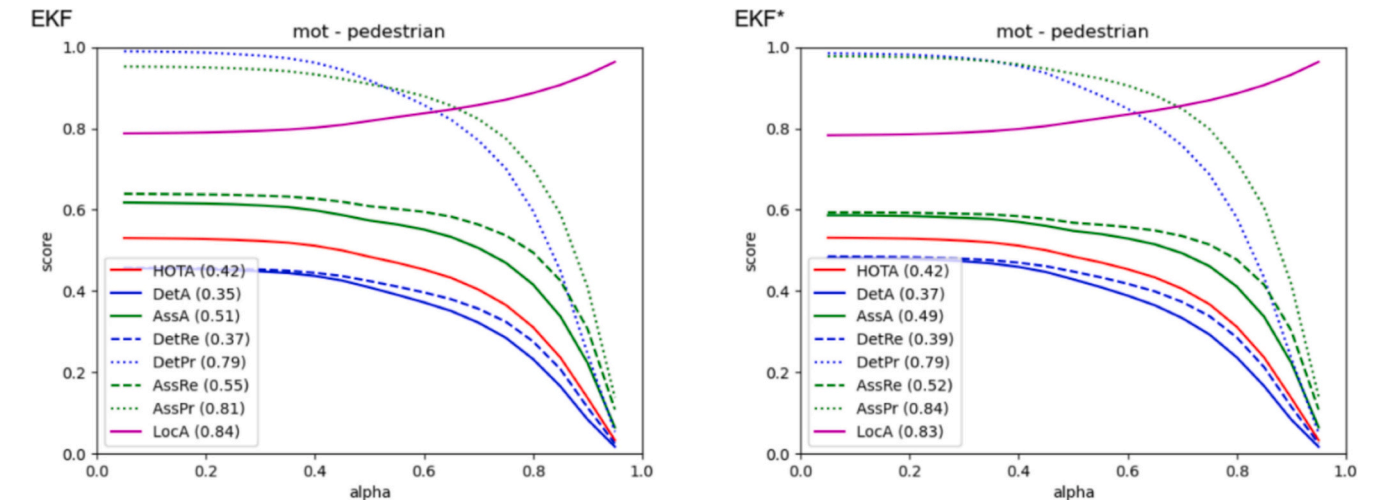


Fig. 14. Pedestrian plot.

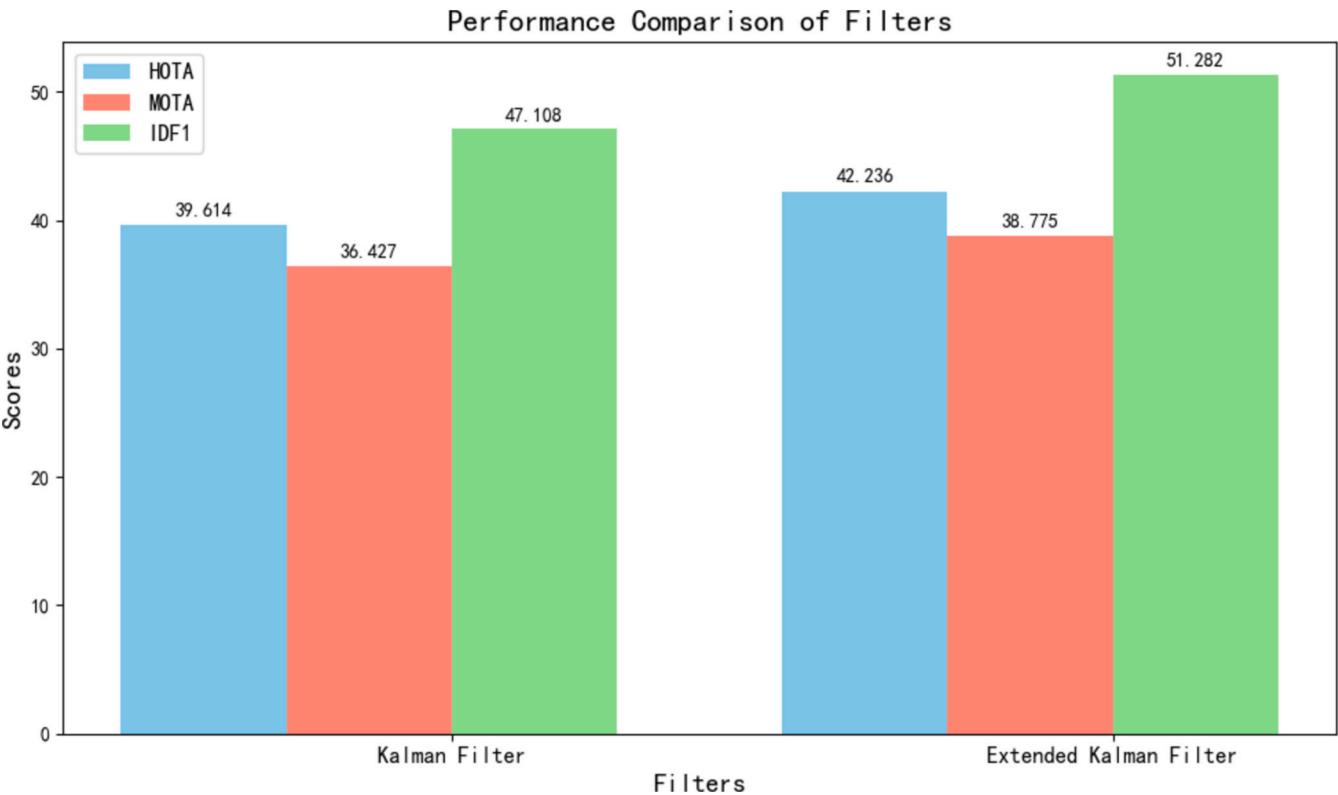


Fig. 15. Performance Comparison of Filters.



Fig. 16. Comparison of Tracking Results Before and After Improved Extended Kalman Filter.

5. Conclusions

This study focuses on the detection and tracking of wildlife, tackling the difficulties of identifying wildlife amidst intricate surroundings. It

introduces an enhanced model, the Yolov8n-DCN-EMGA-ASPFC*. This model integrates data augmentation, Deformable Convolutional Networks version 3 (DCNv3), the Multi-Scale Attention Mechanism (EMGA), and the feature fusion module ASPFC to improve the efficacy of

animal detection and tracking. Specifically, Deformable Convolutional Networks version 3 (DCNv3) substitutes the conventional C2f layer, significantly enhancing the model's ability to extract animal information. The EMGA attention mechanism and the ASPFC module concurrently enhance detection accuracy while maintaining the model's lightweight characteristics. The enhanced Yolov8n-DCN-EMGA-ASPFC* model has elevated the mean average precision (mAP@0.5) by 4.57 %, while accuracy and recall rates have risen by 2.87 % and 4.92 %, respectively. The enhanced Extended Kalman Filter outperforms the original Kalman Filter in Multi-Object Tracking Accuracy (MOTAA) by 3.923 %, Higher-Order Tracking Accuracy (HOTA) by 2.692 %, and IDF1 by 4.246 %.

Furthermore, the research we undertake has significant significance, especially in wildlife conservation, environmental monitoring, and wildlife studies. Improving the accuracy and efficiency of wildlife detection and tracking allows for more precise monitoring of endangered species, enabling conservationists to swiftly evaluate their dynamics and address potential threats, thereby augmenting the precision and scientific rigor of conservation initiatives. This technology facilitates ecosystem health monitoring by employing automated techniques to assess biodiversity alterations and evaluate the effects of environmental changes on animals, thereby offering a scientific basis for environmental protection legislation. Additionally, it can assist researchers in conducting in-depth analyses of wildlife behavior and ecological studies, illuminating critical ecological concerns such as migration patterns and habitat utilization, thereby fostering progress in wildlife conservation and research. This research recognizes limits in dataset processing, including the potential inadequacy of variety and coverage, which may impact the model's generalizability and practical applicability. Future developments entail the continued optimization of dataset construction by incorporating additional samples from other circumstances and wildlife types to improve the model's generalizability and applicability. Future study may investigate more streamlined model structures and methods to enhance the model's real-time performance and computational efficiency, hence increasing its applicability in diverse practical contexts.

Funding

The project was supported by funding from the Youth Scientific Research Foundation of Qinghai University (No.2022-QGY-6).

Institutional review board statement

Not applicable.

Informed consent statement

Not applicable.

CRediT authorship contribution statement

Langkun Jiang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Li Wu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare no conflicts of interest.

Data availability

The data and code supporting the findings of this study are publicly

available. The dataset used in this research can be accessed at <https://www.kaggle.com/datasets/langkunj/wildt-yolov8n-datasets/data>, and the code is available on <https://github.com/Kununk/wildT-YOLOv8n>.

References

- Aharon, N., Orfaig, R., Bobrovsky, B.Z., 2022. BoT-SORT: Robust associations multi-pedestrian tracking[J]. arXiv preprint. arXiv:2206.14651.
- Bernardin, K., Stiefel, R., 2008. Evaluating multiple object tracking performance: the clear mot metrics[J]. Eurasp J. Image Video Process. 2008, 1–10.
- Chandler, R.B., Royle, J.A., 2013. Spatially explicit models for inference about density in unmarked or partially marked populations[J]. Ann. Appl. Stat. 936–954.
- Chen, C., Lu, C.X., Wang, B., et al., 2021. DynaNet: neural Kalman dynamical model for motion estimation and prediction[J]. IEEE Trans. Neural Networks Learn. Syst. 32 (12), 5479–5491.
- Chen, Lingli, et al., 2024. YOLO-SAG: an improved wildlife object detection algorithm based on YOLOv8n. Eco. Inform. 83, 102791.
- Chicco, D., Jurman, G., 2020. The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation[J]. BMC Genomics 21, 1–13.
- Christin, S., Hervet, É., Lecomte, N., 2019. Applications for deep learning in ecology[J]. Methods Ecol. Evol. 10 (10), 1632–1644.
- Cusick, Andrew, et al., 2024. Using machine learning to count Antarctic shag (Leucocarbo bransfieldensis) nests on images captured by remotely piloted aircraft systems. Eco. Inform. 82, 102707.
- Garcia, R.V., Pardal, P.C.P.M., Kuga, H.K., et al., 2019. Nonlinear filtering for sequential spacecraft attitude estimation with real data: cubature Kalman filter, unscented Kalman filter and extended Kalman filter[J]. Adv. Space Res. 63 (2), 1038–1050.
- Gibb, R., Browning, E., Glover-Kapfer, P., et al., 2019. Emerging opportunities and challenges for passive acoustics in ecological assessment and monitoring[J]. Methods Ecol. Evol. 10 (2), 169–185.
- Gor, M., Vora, J., Tanwar, S., et al., 2017. GATA: GPS-Arduino based tracking and alarm system for protection of wildlife animals[C]. In: 2017 International Conference on Computer, Information and Telecommunication Systems (CITS). IEEE, pp. 166–170.
- Hamilton, Grant, et al., 2020. When you can't see the koalas for the trees: using drones and machine learning in complex environments. Biol. Conserv. 247, 108598.
- He, K., Zhang, X., Ren, S., et al., 2015. Spatial pyramid pooling in deep convolutional networks for visual recognition[J]. IEEE Trans. Pattern Anal. Mach. Intell. 37 (9), 1904–1916.
- Hossain, M., Sulaiman, M.N., 2015. A review on evaluation metrics for data classification evaluations[J]. Int. J. Data Min. Knowledge Manag. Proc. 5 (2), 1.
- Hunger, R., 2005. Floating Point Operations in Matrix-Vector calculus[M]. Munich University of Technology, Inst. for Circuit Theory and Signal Processing, Munich, Germany.
- Karp, D., 2020. Detecting small and cryptic animals by combining thermography and a wildlife detection dog[J]. Sci. Rep. 10 (1), 5220.
- Kumar, S., Singh, S.K., 2017. Visual animal biometrics: survey[J]. IET Biomet. 6 (3), 139–156.
- Liu, Y., Shao, Z., Hoffmann, N., 2021. Global attention mechanism: Retain information to enhance channel-spatial interactions[J]. arXiv preprint. arXiv:2112.05561.
- Liu, Yantong, et al., 2024. Camouflage detection: optimization-based computer vision for Alligator sinensis with low detectability in complex wild environments. Eco. Inform. 83, 102802.
- Luiten, J., Osep, A., Dendorfer, P., et al., 2021. Hota: a higher order metric for evaluating multi-object tracking[J]. Int. J. Comput. Vis. 129, 548–578.
- Norouzzadeh, M.S., Nguyen, A., Kosmala, M., et al., 2018. Automatically identifying, counting, and describing wild animals in camera-trap images with deep learning[J]. Proc. Natl. Acad. Sci. 115 (25), E5716–E5725.
- Okafor, E., Pawara, P., Karaaba, F., et al., 2016. Comparative study between deep learning and bag of visual words for wild-animal recognition[C]. In: 2016 IEEE Symposium Series on Computational Intelligence (SSCI). IEEE, pp. 1–8.
- Ouyang, D., He, S., Zhang, G., et al., 2023. Efficient multi-scale attention module with cross-spatial learning[C]. In: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, pp. 1–5.
- Padilla, R., Netto, S.L., Da Silva, E.A.B., 2020. A survey on performance metrics for object-detection algorithms[C]. In: 2020 International Conference On Systems, Signals and Image Processing (IWSSIP). IEEE, pp. 237–242.
- Petso, T., Jamisola Jr., R.S., Mpoeleng, D., 2022. Review on methods used for wildlife species and individual identification[J]. Eur. J. Wildl. Res. 68 (1), 3.
- Pimm, S.L., Alibhai, S., Bergl, R., et al., 2015. Emerging technologies to conserve biodiversity[J]. Trends Ecol. Evol. 30 (11), 685–696.
- Rani, S.V. Jansi, et al., 2024. A novel automated approach for fish biomass estimation in turbid environments through deep learning, object detection, and regression. Eco. Inform. 81, 102663.
- Rawat, U.S., Agarwal, N.K., 2015. Biodiversity: Concept, Threats and Conservation[J]. Redmon, J., Divvala, S., Girshick, R., et al., 2016. You only look once: Unified, real-time object detection[C]. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 779–788.
- Roy, Arunabha M., et al., 2023. WildDect-YOLO: an efficient and robust computer vision-based accurate object localization model for automated endangered wildlife detection. Eco. Inform. 75, 101919.
- Simon, D., 2010. Kalman filtering with state constraints: a survey of linear and nonlinear algorithms[J]. IET Control Theory Appl. 4 (8), 1303–1318.

- Tang, Y., Wang, J., Gao, B., et al., 2016. Large scale semi-supervised object detection using visual and semantic knowledge transfer[C]. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 2119–2128.
- Tharwat, A., 2020. Classification assessment methods[J]. Appl. Comp. Inform. 17 (1), 168–192.
- Trammell, M.J., Oberoi, P., Egenrieder, J., et al., 2019. Contextual label smoothing with a phylogenetic tree on the iNaturalist 2018 challenge dataset[J]. J. Wash. Acad. Sci. 105 (1), 23–46.
- Upreti, G., 2023. Importance of Biodiversity, Ecosystems, and Ecosystem Services[M]// Ecosociocentrism: The Earth First Paradigm for Sustainable Living. Springer Nature Switzerland, Cham, pp. 15–30.
- Urbano, F., Cagnacci, F., Calenge, C., et al., 2010. Wildlife tracking data management: a new vision[J]. Philos. Trans. R. Soc. B 365 (1550), 2177–2185.
- Wagner, Sarah, et al., 2024. Automatic detection of color markings and numbers on trees in point clouds from personal laser scanning (PLS) and terrestrial laser scanning (TLS). Eco. Inform. 82, 102709.
- Wang, C.Y., Bochkovskiy, A., Liao, H.Y.M., 2023a. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors[C]. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 7464–7475.
- Wang, W., Dai, J., Chen, Z., et al., 2023b. InternImage: Exploring large-scale vision foundation models with deformable convolutions[C]. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 14408–14419.
- Wending, Liu, Anqi, Li, Junguo, Zhang, et al., 2018. Automatic identification of terrestrial wildlife in Saihanwula National Nature Reserve based on ROI-CNN [J]. J. Beijing Forest. Univ. 40 (8), 123–131.
- Yacoub, R., Axman, D., 2020. Probabilistic extension of precision, recall, and f1 score for more thorough evaluation of classification models[C]. In: Proceedings of the First Workshop on Evaluation and Comparison of NLP Systems, pp. 79–91.
- Yang, S., Baum, M., 2017. Extended Kalman filter for extended object tracking[C]. In: 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, pp. 4386–4390.
- Zhao, Tingting, et al., 2021. MobileNet-yolo based wildlife detection model: a case study in Yunnan Tongbiguan Nature Reserve, China. J. Intell. Fuzzy Syst. 41 (1), 2171–2181.
- Zhong, Jiageng, et al., 2022. Real-time marine animal detection using yolo-based deep learning networks in the coral reef ecosystem. Int. Arch. Photogramm. Remote. Sens. Spat. Inf. Sci. 46, 301–306.
- Zhu, X., Hu, H., Lin, S., et al., 2019. Deformable convnets v2: More deformable, better results[C]. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 9308–9316.